ENGINEERING ECONOMICS

LECTURE - 14

Inflation

- Persistent increase in the costs of goods and services
- Persistent decrease in buying power of dollar
- inflation rate
  - the % change in prices from one period to another

$$= \frac{\text{price in period \#2 - price in period \#1}}{\text{price in period \#1}} \times 100$$

Example:

Assuming a 3% inflation rate, you would need more money each year to have the same buying power:

- Year 1: $56,000
- Year 2: $57,680
- Year 3: $59,410
- Year 4: $61,193

Your investment would need to earn more than 3% just to beat inflation.

Causes: 1. Cost Push
2. Demand Pull

1. COST PUSH

An increase in costs may lead to an increase in prices.

Examples:
Raw material prices (possibly from abroad) increase...
...Costs to business increase...
...Business still wants to make a profit...
...Business puts its prices up...
...Consumers can buy less with their money...
...Workers demand and receive pay increases...
...Businesses costs increase again...
...Businesses put prices up again

2. DEMAND PULL

- If there is too much demand for goods and services in the economy then prices may be forced upwards.
- Suppliers experience so much demand for their limited number of goods that they decide to put up prices

Deflation in economics is a persistent decrease in the general price level of goods and services, when inflation is below zero percent, resulting in an increase in the real value of money - a negative inflation rate.

When the inflation rate slows down (decreases, but remains positive), this is known as disinflation. It is a substantial drop in the price level.

Depression A time of economic crisis or bad times in commerce, finance, and industry, characterized by falling prices, restriction of credit, low output and investment, many bankruptcies, and a high level of unemployment (many people without jobs). A less severe crisis is usually known as a recession

<u>Types of Capital</u>

Capital is any form of wealth employed to produce more wealth for a firm. It has following three types.

- <u>Fixed</u> - used to purchase the permanent or fixed assets of the business (e.g., buildings, land, equipment, etc.)
- <u>Working</u> - used to support the small company's normal short-term operations (e.g., buy inventory, pay bills, wages, salaries, etc.)
- <u>Growth</u> - used to help the small business expand or change its primary direction.

Equity and Debt Capital

- **Equity** represents the personal investment of the owner(s) in the business.
- Does *not* have to be repaid with interest like a loan does.

- **Debt Capital** is the capital that a business raises by taking out a loan.
- Must be repaid with interest at some future date.

Bond A certificate of debt issued by a government or corporation guaranteeing payment of the original investment plus interest by a specified future date.

Credit: **The giving of goods and services in return for the promise of payment at a future time. The payment usually has interest attached.**

Dividend: Profits of a firm that are distributed or given out to its investors (stockholders).

Tax: A contribution for the support of a government required of persons, groups, or businesses. There are many different kinds of taxes including income, sales, state. local, federal taxes.

Tariff: A tax on imports or exports by the government.

1. **Income taxes**
2. **Property taxes**
3. **Sales taxes**
4. **Excise taxes**

Discussion